CO1- Assessment Tool1 Short Answer Test

DSA02 - COMPUTER VISON

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1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Answer:

Sampling determines the number of pixels used to represent an image. If the sampling rate is too low, important image details are lost, resulting in a blurred image. This happens because fewer pixels cannot accurately represent fine details.

Correction:

- Increase the sampling rate by using a higher-resolution camera.
- Apply image sharpening or super-resolution techniques to improve clarity.
- Ensure proper focus during image capture.

2. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Answer:

- **Pixel Resolution (Spatial Resolution):** Refers to the number of pixels in an image. Higher pixel resolution provides finer details, making it easier to detect small abnormalities such as tumors.
- **Intensity Resolution (Gray-Level Resolution):** Refers to the number of brightness levels available. Higher intensity resolution allows better differentiation between tissues with similar brightness.

In Medical Imaging:

- High pixel resolution improves anatomical detail.
- High intensity resolution improves contrast between healthy and diseased tissues.

3. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Answer:

Image formation models describe how light from a scene is captured by a camera sensor. These models consider illumination, object reflectance, camera optics, and sensor characteristics.

In autonomous driving, image formation models help by:

- Correcting lighting variations.

- Reducing shadows and glare.
- Improving visibility in rain or fog.
- Producing clearer images for lane detection and obstacle recognition.

This results in safer and more accurate driving decisions.

4. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Answer:

Noise is introduced during image acquisition due to:

- Low lighting conditions.
- Electronic sensor noise.
- High ISO settings.
- Thermal effects inside the camera.

Noise can be minimized by:

- Using better-quality image sensors.
- Increasing illumination.
- Applying image filtering techniques such as Median Filter or Gaussian Filter.
- Using appropriate exposure settings.

These methods improve image clarity and accuracy.

5. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Answer:

Facial recognition involves multiple Computer Vision levels:

1. **Image Acquisition:** Capture the face using a camera.
2. **Preprocessing:** Remove noise and normalize lighting.
3. **Feature Extraction:** Identify facial landmarks such as eyes, nose, and mouth.
4. **Recognition:** Compare extracted features with stored facial data.
5. **Decision Making:** Verify or identify the person's identity.

Each stage contributes to accurate recognition.

6. Compare two image acquisition systems (CCD vs CMOS) in terms of their impact on image quality in practical applications.

Answer:

CCD Sensor	**CMOS Sensor**
Better image quality	Good image quality
Lower noise	Slightly higher noise
Higher power consumption	Lower power consumption
Slower image capture	Faster image capture
Used in scientific and medical imaging	Used in smartphones, surveillance cameras, and autonomous vehicles

Conclusion: CCD provides higher image quality, while CMOS offers faster performance and lower power usage.

7. A digital camera produces images with banding effects. Analyze how quantization levels contribute to this issue and propose a solution.

Answer:

Banding occurs when the number of intensity levels is too low. Smooth brightness transitions are replaced with visible intensity steps due to coarse quantization.

Solution:

- Increase the bit depth (e.g., from 8-bit to 12-bit or 16-bit).
- Use higher quantization levels.
- Apply dithering techniques to smooth intensity transitions.

This produces more natural-looking images.

8. A surveillance image appears noisy in low light. Analyze how sensing and acquisition conditions contribute to noise.

Answer:

Low-light conditions reduce the amount of light reaching the sensor. To compensate, the camera increases sensor gain (ISO), which amplifies electronic noise.

Factors contributing to noise:

- Poor illumination.
- Small image sensor.
- High ISO settings.

- Long exposure times.

Solutions:

- Improve lighting.
- Use larger sensors.
- Apply noise reduction algorithms.
- Use cameras designed for low-light imaging.

9. In medical imaging, explain how increasing the sampling rate improves diagnostic accuracy.

Answer:

Increasing the sampling rate increases the number of pixels used to represent an image.

Benefits include:

- Better visualization of small anatomical structures.
- Reduced aliasing effects.
- Improved edge definition.
- Higher diagnostic accuracy for detecting diseases.

For example, high-resolution MRI or CT scans help doctors identify small tumors that may not be visible at lower resolutions.

10. In a facial recognition system, explain how resolution impacts feature extraction accuracy.

Answer:

Image resolution determines the amount of facial detail available for analysis.

Higher resolution:

- Captures finer facial features.
- Improves landmark detection.
- Increases recognition accuracy.
- Reduces false matches.

Lower resolution:

- Loses important facial details.
- Makes feature extraction difficult.
- Decreases recognition performance.

Therefore, high-resolution images are essential for reliable facial recognition systems.